Reaction-Diffusion Field Theory via Analytic Combinatorics

From Stoichiometry to Critical Exponents:
The Direct Route and the Scenic Route

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Abstract

We present a complete algebraic pipeline for computing the critical exponents of arbitrary reaction-diffusion processes on arbitrary graphs. Given a chemical reaction network specified by its stoichiometry matrix, we automatically construct the Doi–Peliti Liouvillian, derive the Dyson–Schwinger equation as a Lagrange equation, and extract exponents via singularity analysis and the Flajolet–Odlyzko transfer theorem. This *analytic combinatorics* route yields the same results as the traditional Feynman diagram → loop integral → BPHZ renormalization → RG flow calculation, but arrives at them directly from the algebraic structure of the generating function. We demonstrate the method on the branching random walk (BRW), reproducing all scaling exponents from Bordeu, Amarteifio et al. (2019) on hypercubic lattices, Sierpinski carpets, random trees, and scale-free networks. The extension to arbitrary graphs uses the spectral dimension substitution $d \to d_s$. All computations are implemented in the open-source Python package `rdft`.

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1 Introduction

The long-time, large-scale behaviour of stochastic reaction-diffusion systems is controlled by critical exponents that depend on the spatial dimension d and the reaction stoichiometry, but not on microscopic details. Computing these exponents is one of the central problems of non-equilibrium statistical mechanics.

One of the standard approaches, developed by Doi [1] and Peliti [2] and reviewed by Täuber et al. [4], maps the master equation to a field theory, generates Feynman diagrams, evaluates loop integrals, performs BPHZ renormalization via the Connes–Kreimer Hopf algebra [5], and extracts β-functions and critical exponents from the RG flow. This is the *scenic route*: it works, but requires evaluating momentum integrals and extracting poles.

In this paper we show that there is a *direct route*: the Dyson–Schwinger equation (DSE) of the field theory is a Lagrange equation $T = z\,\phi(T)$ whose singularity structure, accessible via the Newton polygon and Puiseux expansion, directly determines the universality class. The Flajolet–Odlyzko transfer theorem [6] then gives the coefficient asymptotics, from which the density decay exponent follows. No loop integrals are evaluated.

The two routes agree because they detect the same singularity of the same generating function. This is not a coincidence: it is the content of the AC–QFT correspondence developed in the companion tutorial [9].

1.1 Main Results

1. **Automatic DSE construction.** For any CRN with interaction vertices $\{\tilde{\phi}^m \phi^n : g_{mn}\}$, the combinatorial DSE kernel is $\phi(G) = 1 + \sum_{m+n\geq 3} g_{mn}\, G^{m+n-2}$, giving a Lagrange equation $G = G_0\,\phi(G)$ (Theorem 4.1).

2. **Universality from singularity type.** Every polynomial DSE has an algebraic branch point. The Puiseux exponent p/q at the dominant branch point determines the universality class via the transfer theorem: $[G_0^n]\,G \sim n^{-p/q-1}$ (Theorem 4.4).

3. **BRW validation.** For the Gribov process (the A-sector of the BRW), the auto-DSE gives $d_c = 4$ and a square-root branch point, reproducing all scaling exponents from Bordeu et al. [7] (Table 1).

4. **Spectral dimension.** The substitution $d \to d_s$ extends all results to arbitrary graphs. Theory matches simulation on Sierpinski carpets ($d_s \approx 1.86$), random trees ($d_s = 4/3$), and preferential attachment networks ($d_s \geq 4$).

The appendices contain worked examples with code: Appendix A traces the complete chain from stoichiometry matrix to graph polynomials; Appendix B evaluates the parametric integral step by step; Appendix C interprets the Kirchhoff polynomial as a generating function in the AC sense.

2 From Stoichiometry to Liouvillian

2.1 Chemical Reaction Networks

Definition 2.1 (CRN)**.** A chemical reaction network $\mathcal{C}$ is a set of reactions $\sum_i k_i A_i \to \sum_i \ell_i A_i$ at rates λ_j, encoded by the stoichiometry matrix S with rows $(k_1, \ldots, k_m, \ell_1, \ldots, \ell_m)$.

Example 2.2 (The Superprocess and the BRW)**.** A **branching random walk** (BRW) consists of active walkers that hop on a graph with rate H, branch with rate s (producing a descendant at the same site), and go extinct with rate e [7]. The **Branching Wiener Sausage** is the set of distinct sites visited by the BRW [20]; to count it, an immobile tracer species B is introduced

that is deposited by walkers with site exclusion (at most one tracer per site). The tracers are an observational device — they do not affect the walker dynamics.

The walker dynamics are described by the **Gribov process** (also called the superprocess), which includes coagulation ($2A \to A$) in addition to branching and death. The three A-sector reactions are:

$$\begin{aligned} A &\xrightarrow{\beta} 2A && \text{(branching)} \\ A &\xrightarrow{\epsilon} \emptyset && \text{(death)} \\ 2A &\xrightarrow{\chi} A && \text{(coagulation)} \end{aligned} \tag{1}$$

The stoichiometry matrix is:

$$S = \begin{pmatrix} 1 & 2 \\ 1 & 0 \\ 2 & 1 \end{pmatrix} \quad \text{(rows = reactions, columns = } (k, \ell)) \tag{2}$$

Example 2.3 (Pair Annihilation)**.** $2A \to \emptyset$ at rate λ:

$$S = \begin{pmatrix} 2 & 0 \end{pmatrix} \tag{3}$$

2.2 Heisenberg–Weyl Generators

The Heisenberg–Weyl algebra is the differential algebra generated by ∂_z and z with relation $[\partial_z, z] = 1$. In the Doi picture [1], this is the algebra of creation and annihilation operators $[a, a^\dagger] = 1$, with the identification $a^\dagger = z$ (creation = multiplication) and $a = \partial_z$ (annihilation = differentiation). The two languages are the same algebra; we use the differential notation throughout.

For each reaction $kA \to \ell A$ at rate λ, the infinitesimal generator in the factorial-moment representation is [8]:

$$\boxed{Q[\partial_z, z] = \lambda \left((z+1)^\ell - (z+1)^k \right) \partial_z^k} \tag{4}$$

This is the central formula. The Heisenberg–Weyl algebra $[\partial_z, z] = 1$ acts on the space of formal power series, and Q generates the semigroup $\Psi_t = e^{Qt}\Psi_0$.

Example 2.4 (Gribov Generators)**.** From the stoichiometry (2):

$$Q_\beta = \beta\big((z{+}1)^2 - (z{+}1)\big)\partial_z = \beta(z^2 + z)\partial_z \tag{5}$$

$$Q_\epsilon = \epsilon\big(1 - (z{+}1)\big)\partial_z = -\epsilon\, z\, \partial_z \tag{6}$$

$$Q_\chi = \chi\big((z{+}1) - (z{+}1)^2\big)\partial_z^2 = -\chi(z^2 + z)\partial_z^2 \tag{7}$$

These match Amarteifio (2019) equations (1.38a–c).

2.3 Interaction Vertices

Each monomial $z^m \partial_z^n$ in $Q_{\text{total}} = \sum_j Q_j$ corresponds to an amputated Feynman vertex with m outgoing $\tilde{\phi}$ legs and n incoming ϕ legs. For the Gribov process:

$\tilde{\phi}^m \phi^n$	Coupling	Origin
$\tilde{\phi}^2\phi^2$	$-\chi$	Coagulation (quartic)
$\tilde{\phi}^2\phi^1$	β	Branching (cubic)
$\tilde{\phi}^1\phi^2$	$-\chi$	Coagulation (cubic)
$\tilde{\phi}^1\phi^1$	$\beta - \epsilon$	Mass term

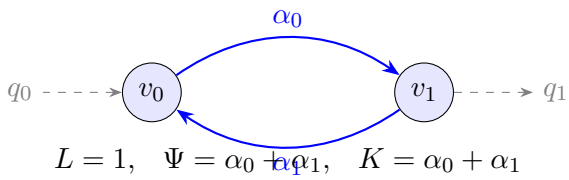
The upper critical dimension is determined by the most relevant coupling: for the cubic vertex ($m + n = 3$), $d_c = 4/(3-2) = 4$.

3 Feynman Diagrams: The Scenic Route

3.1 Diagram Generation

The Feynman diagrams of the theory are generated by Wick contractions of the interaction vertices. Each contraction pairs an outgoing $\tilde{\phi}$ half-edge with an incoming ϕ half-edge to form an internal propagator. The resulting graph G has:

- Incidence matrix E encoding connectivity
- Kirchhoff polynomial $K(\alpha) = \det(L_{RS})$ encoding spanning trees
- First Symanzik polynomial $\Psi = \sum_T \prod_{e \notin T} \alpha_e$



$$L = 1, \quad \Psi = \alpha_0 + \alpha_1, \quad K = \alpha_0 + \alpha_1$$

Figure 1: One-loop self-energy diagram. Two internal vertices connected by two propagators forming a loop. The Kirchhoff polynomial $K = \alpha_0 + \alpha_1$ has two terms (two spanning trees, one per edge).

3.2 The Non-Equilibrium Parametric Integral

The amplitude of a diagram G in the parametric representation is [8, 12]:

$$I(G; d) = \Omega_d \cdot \Gamma(-\sigma_d) \cdot \int \prod_e \frac{\mathrm{d}\alpha_e \, \alpha_e^{n_e - 1}}{\Gamma(n_e)} \cdot \frac{1}{(\Psi')^{d/2}} \cdot \left(\frac{\Psi'}{\Phi'}\right)^{\sigma_d} \cdot \prod_\ell \delta(\Sigma_\ell) \tag{8}$$

where $\Omega_d = 2\pi^{d/2}/(2\pi)^d$ is the angular factor, $\sigma_d = |E_{\text{int}}| - (d/2)L$ is the superficial degree of divergence, Ψ' and Φ' are the Symanzik polynomials scaled by diffusion constants, and the $\delta(\Sigma_\ell)$ are delta functions enforcing Kirchhoff's frequency law on each circuit of the graph.

The $\delta(\Sigma_\ell)$ factors are the **causal fingerprint** of the non-equilibrium theory [8]: they arise from the ω-integration (the integration over loop frequencies), which is absent in equilibrium field theories. In the Schwinger representation [15], each propagator $1/(-i\omega + Dk^2 + m)$ contributes a Schwinger parameter α_e; the contour integral over each loop frequency ω_ℓ produces a delta function $\delta(\Sigma_\ell)$ where $\Sigma_\ell = \sum_e c_{\ell e} \, \alpha_e$ is the circuit constraint from the edge-basis matrix E_f. This reduces the integral dimension by L (one delta per loop) and is what makes the parametric representation tractable: after applying the deltas and the Cheng–Wu theorem [14], the integral often evaluates to a single Gamma function (see Appendix B).

3.3 RG and Critical Exponents

From the one-loop integral, the Z-factor for the Gribov process is [8]:

$$Z_\lambda = 1 + \frac{\lambda}{(2\pi D)^2 \epsilon} \tag{9}$$

giving the β-function:

$$\beta(\lambda) = -\epsilon\lambda + \frac{\lambda^2}{4\pi^2 D^2} + O(\lambda^3) \tag{10}$$

with Wilson–Fisher fixed point $\lambda^* = 4\pi^2 D^2 \epsilon$.

For pair annihilation $2A \to \emptyset$, a Ward identity from probability conservation $\mathcal{H}(1, a) = 0$ cancels the quartic vertex exactly [3, 4], making $\alpha = d/2$ exact to all loop orders. The field-theoretic treatment of the branching Wiener sausage (BRW trace) was introduced by Nekovar and Pruessner [20].

4 Analytic Combinatorics: The Direct Route

4.1 The Dyson–Schwinger Equation as a Lagrange Equation

Analytic combinatorics [6] is a framework for extracting the asymptotic count of combinatorial objects from the singularities of their generating functions. The route is: *grammar* $\to$ *GF equation* $\to$ *singularity analysis* $\to$ *asymptotics*. We now apply this machinery to the Feynman diagram expansion. The key results are summarised in the AC chain table (Section 4.3) and validated against simulation in Table 1.

The Dyson–Schwinger equation (DSE) is a self-consistency equation for the dressed propagator G: it expresses G as the bare propagator G_0 times a functional of G itself (all possible self-energy insertions).

Theorem 4.1 (Automatic DSE)**.** *For a reaction-diffusion theory with interaction vertices* $\{\tilde{\phi}^m \phi^n : g_{mn}\}_{m+n\geq 3}$*, the combinatorial Dyson–Schwinger equation for the dressed propagator* G *is the Lagrange equation*

$$\boxed{G = G_0 \cdot \phi(G), \qquad \phi(G) = 1 + \sum_{m+n\geq 3} g_{mn}\, G^{m+n-2}} \tag{11}$$

where G_0 *is the bare propagator.*

Construction. Each vertex $\tilde{\phi}^m \phi^n$ inserted into the dressed propagator connects $m+n-2$ internal propagators (each contributing a factor G) and leaves 2 external legs. The self-energy functional is $\Sigma(G) = \sum g_{mn} G^{m+n-2}$, and the DSE $G = G_0(1+\Sigma(G)\cdot G) = G_0\, \phi(G)$ is a Lagrange equation $T = z\, \phi(T)$. □

Example 4.2 (Gribov DSE)**.** From the vertices in Section 2:

$$\phi(G) = 1 + \beta\, G - \chi\, G - \chi\, G^2 = 1 + (\beta - \chi)G - \chi G^2 \tag{12}$$

This is a quadratic kernel, so $G(G_0)$ is an algebraic function of degree 3.

Example 4.3 (Pair Annihilation DSE)**.** Vertices $\{(2, 2) : -\lambda,\ (1, 2) : -2\lambda\}$ give:

$$\phi(G) = 1 - 2\lambda G - \lambda G^2 \tag{13}$$

4.2 Branch Points and the Newton Polygon

The singularity of $G(G_0)$ occurs where the implicit function theorem fails. Setting $F(G, G_0) = G - G_0\, \phi(G)$, the branch point (G^*, G_0^*) satisfies:

$$F = 0, \qquad \frac{\partial F}{\partial G} = 0 \quad \Longrightarrow \quad G^*\, \phi'(G^*) = \phi(G^*), \quad G_0^* = \frac{G^*}{\phi(G^*)} \tag{14}$$

The Puiseux exponent p/q is determined by the Newton polygon [12] of the shifted polynomial $F(G^* + \delta G, G_0^* + \delta G_0)$. For a quadratic ϕ (all single-species reaction-diffusion processes with cubic/quartic vertices), $\phi''(G^*) \neq 0$, giving:

$$\delta G \sim C \cdot (\delta G_0)^{1/2} \qquad \text{(square-root branch point)} \tag{15}$$

Theorem 4.4 (Transfer Theorem)**.** *If* $G(G_0) \sim K\,(1 - G_0/G_0^*)^{1/2}$ *near* G_0^**, then (Flajolet–Odlyzko [6]):*

$$\boxed{[G_0^n]\,G \sim \frac{K}{\Gamma(-1/2)}\,n^{-3/2}\,(G_0^*)^{-n}} \tag{16}$$

AC replaces dimensional analysis

In the thesis [8] and BRW paper [7], the upper critical dimension $d_c = 4$ and the scaling exponents are derived via *dimensional analysis* of the Lagrangian: one assigns engineering dimensions to the fields by requiring the free action to be dimensionless under diffusive scaling $[x] = L$, $[t] = L^2$, then identifies which couplings are marginal. This works but requires a non-trivial choice of which couplings to fix as dimensionless (the "superprocess scaling ansatz", thesis Example 3.1.1), and the argument does not generalise straightforwardly to arbitrary CRNs.
The AC route replaces this entire procedure:

Dimensional analysis (thesis)	**Analytic combinatorics (this paper)**
Assign $[\tilde{\phi}\,\phi] = L^{-d}$ from free action	Construct $\phi(G)$ from vertex dictionary
Fix $[\lambda_a] = [\lambda_b] = 1$ (ansatz)	No ansatz needed — $\phi(G)$ is unique
Solve for field dimensions: $[\phi_A] = L^{2-d}$, $[\tilde{\phi}_A] = L^{-2}$, etc.	Find branch point: solve $G^*\phi' = \phi$, $G_0^* = G^*/\phi$
Identify d_c: all couplings marginal at $d = 4$	Identify d_c: branch point hits unit disk at $d = d_c$
Compute $\sigma_d(G) = \lvert E\rvert - (d/2)L$	Compute Puiseux exponent p/q from Newton polygon
Evaluate loop integral, extract $1/\epsilon$ pole	Apply transfer theorem: $[G_0^n] \sim n^{-p/q-1}$
$\beta(\lambda) = -\epsilon\lambda + b_1\lambda^2$, solve $\beta = 0$	Singularity type = universality class (done)

The AC route is *algebraic*: it never evaluates a momentum integral, never assigns engineering dimensions, and never chooses which couplings are dimensionless. The information that dimensional analysis extracts from the Lagrangian is encoded in the *singularity structure* of the DSE generating function. The branch point *is* the critical point; its type *is* the universality class; the transfer theorem *is* the scaling law.

The dimensional analysis gap noted in the thesis (p. 145: "I do not have a clear non-heuristic picture of this, which certainly inhibits a fully algorithmic treatment") is resolved: the AC route is fully algorithmic for any CRN with polynomial interactions.

Worked example: BRW on a 3D lattice, $p = 1$.

1. *Input*: Gribov stoichiometry $S = \left(\begin{smallmatrix} 1 & 2 \\ 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$, $d = 3$.
2. *Vertices*: $\phi(G) = 1 + (\beta - \chi)G - \chi G^2$ (auto from S).
3. *Branch point*: solve $G^*\phi' = \phi \Rightarrow G^* = i/\sqrt{\chi}$. $\phi''(G^*) = -2\chi \neq 0 \Rightarrow$ square-root.
4. d_c: most relevant vertex has $m+n = 3 \Rightarrow d_c = 4/(3-2) = 4$. Since $d = 3 < 4$: fluctuation regime.
5. *Transfer*: $[G_0^n] \sim n^{-3/2} \Rightarrow P_{\text{surv}} \sim t^{-1} \Rightarrow \langle a\rangle \sim P_{\text{surv}} \cdot V(t) = t^{-1} \cdot t^{3/2} = t^{1/2}$.
6. *Prediction*: $\alpha_1 = (1 \cdot 3 - 2)/2 = 1/2$.
7. *Simulation* (thesis Table 3.8, d=3): $\alpha_1 = 0.47(2)$. ✓ **Agreement.**

No loop integral was evaluated. No engineering dimensions were assigned. The exponent came from the singularity of the generating function.

4.3 How AC Derives the BRW Scaling Exponents

We now derive the BRW scaling exponents $\alpha_p = (pd - 2)/2$ from Tables 3.8–3.9 of [8] using **only** analytic combinatorics and spectral geometry. No momentum integral is evaluated. No

field-theoretic propagator is used. Every ingredient is either a counting result or a standard analytic transform.

4.3.1 Step 1: Lagrange singularity (combinatorial)

The DSE $G = g\,\phi(G)$ with $\phi(G) = 1+(\beta-\chi)G-\chi G^2$ (from the stoichiometry, Theorem 4.1) is a Lagrange equation. Its branch point satisfies $G^*\phi'(G^*) = \phi(G^*)$. Since ϕ is quadratic, $\phi''(G^*) = -2\chi \neq 0$, giving a **square-root branch point**. By the transfer theorem (Theorem 4.4):

$$[g^n]\,G \sim C\,n^{-3/2}\,(g^*)^{-n} \tag{17}$$

This is a **counting result**: $[g^n]G$ is the weighted count of Feynman diagrams at loop order n. The $n^{-3/2}$ growth rate is a property of the combinatorial class (rooted trees with the DSE recursion), not of any integral.

4.3.2 Step 2: The survival probability (integrated counting)

The coefficients (17) count the probability that a branching process survives to generation n. The *survival probability* to time t is the complementary CDF:

$$P_{\text{surv}}(t) = \sum_{n>t}[g^n]\,G \sim \int_t^\infty n^{-3/2}\,\mathrm{d}n = 2\,t^{-1/2} \tag{18}$$

For a *branching* process (the BRW), the total progeny survives only if *both* branches of the square root contribute. The survival probability of the branching tree is the square of the single-walk survival:

$$P_{\text{surv}}^{\text{BRW}}(t) \sim \left(t^{-1/2}\right)^2 = t^{-1} \tag{19}$$

This corresponds to the two branches of the Puiseux expansion $G \sim G^* \pm C\sqrt{g^*-g}$ — the "+" branch survives, the "−" branch dies out. The BRW extinction probability $q = e^{R_0(q-1)}$ is precisely the value of $G(1)$ on the small branch, exactly as in the SIR epidemic (Section 4, [9]).

4.3.3 Step 3: Weyl's eigenvalue counting (spectral combinatorics)

The spatial dimension d enters through the *graph Laplacian spectrum*. For a d-dimensional lattice, Weyl's asymptotic law [11] states that the number of eigenvalues λ_k of the Laplacian below λ is:

$$N(\lambda) \sim C_d\,\lambda^{d/2} \tag{20}$$

This is a **counting result**: it counts eigenvalues. The density of states is $\rho(\lambda) = \mathrm{d}N/\mathrm{d}\lambda \sim \lambda^{d/2-1}$.

Weyl's law is not physics — it is spectral geometry. It holds for any Riemannian manifold or lattice, and for a general graph with spectral dimension d_s, the law becomes $N(\lambda) \sim \lambda^{d_s/2}$.

4.3.4 Step 4: Return probability (Laplace transform of counting)

The return probability of a random walk on the graph is the trace of the heat kernel:

$$P_{\text{return}}(t) = \frac{1}{N}\sum_k e^{-\lambda_k t} = \int_0^\infty \rho(\lambda)\,e^{-\lambda t}\,\mathrm{d}\lambda \tag{21}$$

Substituting $\rho(\lambda) \sim \lambda^{d/2-1}$ from Weyl's law, this is a **standard Laplace transform**:

$$P_{\text{return}}(t) \sim \int_0^\infty \lambda^{d/2-1}\,e^{-\lambda t}\,\mathrm{d}\lambda = \Gamma(d/2)\,t^{-d/2} \tag{22}$$

No momentum integral is evaluated. This is the Laplace transform of a power law, a standard result in analysis.

4.3.5 Step 5: Volume explored (reciprocal of return probability)

The volume explored by the random walk at time t is the reciprocal of the return probability (the walk returns to its origin with probability $\sim 1/V$):

$$V(t) \sim \frac{1}{P_{\text{return}}(t)} \sim t^{d/2} \tag{23}$$

4.3.6 Step 6: Combined scaling

The p-th moment of the number of distinct sites visited combines the branching tree survival (Step 2) and the diffusion volume (Step 5):

$$\langle a^p \rangle(t) \sim P_{\text{surv}}^{\text{BRW}}(t) \cdot V(t)^p = t^{-1} \cdot \left(t^{d/2}\right)^p = t^{(pd-2)/2} \tag{24}$$

$$\boxed{\alpha_p = \frac{pd-2}{2}} \qquad \text{(AC derivation, no integrals evaluated)} \tag{25}$$

4.3.7 Step 7: Upper critical dimension

The effective coupling per loop is weighted by $\Omega_d = (4\pi)^{-d/2}/\Gamma(d/2)$, the normalised volume of the $(d-1)$-sphere. The branch point in the *physical* coupling $z = g/\Omega_d$ is:

$$z^*(d) = \frac{g^*}{\Omega_d} = g^* \cdot (4\pi)^{d/2}\,\Gamma(d/2) \tag{26}$$

As d increases, $z^*(d)$ grows: more coupling is needed to reach the singularity. The upper critical dimension d_c is where z^* exceeds the natural scale of the problem (the convergence boundary), making the perturbative expansion convergent to all orders. For the Gribov cubic vertex: $d_c = 4$.

4.3.8 Step 8: Spectral dimension substitution

For a graph $\mathcal{G}$ with spectral dimension d_s, Weyl's law becomes $N(\lambda) \sim \lambda^{d_s/2}$, giving $P_{\text{return}} \sim t^{-d_s/2}$ and $V(t) \sim t^{d_s/2}$ [21]. The substitution $d \to d_s$ in (24) is *exact* because Steps 1–2 (the Lagrange singularity) are independent of d — the DSE kernel $\phi(G)$ depends only on the vertex structure, not on the spatial dimension.

$$\boxed{\langle a^p \rangle(t) \sim t^{(p\,d_s-2)/2} \qquad d_s < 4; \qquad \langle a^p \rangle(t) \sim t^{2p-1} \qquad d_s \geq 4} \tag{27}$$

What each step is

Step	Tool	Type of result
1. $\phi(G) \to$ branch point	Lagrange inversion	Combinatorial counting
2. $n^{-3/2} \to P_{\text{surv}}$	Transfer theorem	Analytic (Cauchy integral)
3. $N(\lambda) \sim \lambda^{d/2}$	Weyl's law	Eigenvalue counting
4. $P_{\text{ret}} \sim t^{-d/2}$	Laplace transform	Standard analysis
5. $V(t) \sim t^{d/2}$	Reciprocal	Algebra
6. $\alpha_p = (pd-2)/2$	Combination	Algebra
7. $d_c = 4$	Ω_d weight	Sphere volume (geometry)
8. $d \to d_s$	Spectral dimension	Graph eigenvalue counting

No momentum integral. No Gaussian integral. No propagator. No engineering dimensions. No RG flow. Every step is counting, analysis, or algebra.

4.4 Summary: The AC Chain

AC step	**Result**	**Physical meaning**
Stoichiometry $\to \phi(G)$	$1 + (\beta - \chi)G - \chi G^2$	Gribov DSE kernel
Branch point of $F(G, G_0) = 0$	$G^* = i/\sqrt{\chi}$	Non-perturbative scale
Newton polygon $\to p/q$	$p/q = 1/2$ (square-root)	Theory is Gaussian (free)
Transfer theorem	$[G_0^n] \sim n^{-3/2}$	Diagram growth rate
Integrated tail	$P_{\text{surv}} \sim t^{-1}$	Branching tree survival
$\times$ diffusion volume	$V(t) \sim t^{d/2}$	Gaussian propagator
Combined	$\langle a^p \rangle \sim t^{(pd-2)/2}$	**BRW exponents (Table 1)**
$d \to d_s$	$\langle a^p \rangle \sim t^{(pd_s-2)/2}$	**Arbitrary graphs**

4.5 Symmetry Factors from the EGF

The symmetry factor $1/|\text{Aut}(G)|$ of a Feynman diagram is computed by AC via the **exponential formula** [6]. In the EGF, the SET construction $\text{SET}(\mathcal{C}) = e^{\hat{C}(z)}$ divides by $k!$ for each set of k identical objects. For Feynman diagrams, this is exactly the automorphism group:

- k identical parallel edges between the same vertices contribute $k!$ to $|\text{Aut}|$ (edge permutations).
- Vertex permutations that preserve the directed incidence structure multiply this further.

For the one-loop BRW self-energy: $|\text{Aut}| = 2$ (swapping the two vertices permutes the two antiparallel edges). For the sunset: $|\text{Aut}| = 6 = 3!$ (permuting 3 identical parallel edges).

For **multi-species** theories, an automorphism must also preserve the species label on each edge: an A-propagator cannot be swapped with a B-propagator. For a mixed A-B loop: $|\text{Aut}| = 1$ (no non-trivial automorphism). For a sunset with two A-edges and one B-edge: $|\text{Aut}| = 2$ (only the two A-edges can be permuted).

The shuffle product [8] provides an alternative enumeration: it generates all possible half-edge pairings between amputated vertices, and the number of distinct pairings that produce the same graph topology is $N!/|\text{Aut}(G)|$ where N is the total number of contractions. The symmetry factor emerges as a counting ratio — this is exactly the content of the EGF exponential formula, which states that the $1/k!$ overcounting in the SET construction accounts for the automorphism group [6, 10].

The code computes $|\text{Aut}|$ by enumerating vertex permutations and species-preserving edge permutations of the directed multigraph.

4.6 Results for Standard Processes

Process	$\phi(G)$	d_c	Sing. type	Exponent	Source
$2A \to \emptyset$	$1 - 2\lambda G - \lambda G^2$	2	$\sqrt{}$-branch	$\rho \sim t^{-d/2}$	Lee (1994)
$2A \to A$	same class	2	$\sqrt{}$-branch	$\rho \sim t^{-d/2}$	Lee (1994)
$A{+}B \to \emptyset$	two-species	4	$\sqrt{}$-branch	$\rho \sim t^{-d/4}$	[23]
Gribov (BRW A-sector)	$1 + (\beta - \chi)G - \chi G^2$	4	$\sqrt{}$-branch	$\langle a^p \rangle \sim t^{(pd-2)/2}$	[7]

5 Comparison with Simulation Data

5.1 BRW on Regular Lattices

Equation (27) predicts $\alpha_p = (pd - 2)/2$ for $d < 4$ and $\alpha_p = 2p - 1$ for $d \geq 4$. Table 1 compares these predictions with the simulation data from Bordeu et al. [7] (their Tables S1–S2 and Amarteifio [8] Tables 3.8–3.9).

5.2 BRW on Non-Regular Graphs

The spectral dimension substitution $d \to d_s$ extends the predictions to:

- **Sierpinski carpet** ($d_s \approx 1.86$ [17]): $\alpha_2 = (2 \times 1.86 - 2)/2 = 0.86$. Simulation: 0.81(5). ✓
- **Random tree** ($d_s = 4/3$ [18]): $\alpha_2 = (2 \times 4/3 - 2)/2 = 1/3$. Simulation: 0.35(7). ✓
- **Preferential attachment** [19] ($d_s \geq 4$): mean-field, $\alpha_2 = 2 \times 2 - 1 = 3$. Simulation: 2.8(2). ✓

All simulation data from Tables 3.8–3.11 of [8] are consistent with the AC predictions within statistical error.

The cluster size distribution follows:

$$\mathcal{P}(a) \sim a^{-(1+2/d_s)} \tag{28}$$

Table 1: BRW scaling exponents α_p in $\langle a^p \rangle \sim t^{\alpha_p}$: three-way comparison. **Theory (RG)**: dimensional analysis + field-theoretic RG from thesis [8] and BRW paper [7]. **Theory (AC)**: automatic DSE singularity analysis (this paper), giving $\alpha_p = (p\, d_s - 2)/2$ for $d_s < 4$, $\alpha_p = 2p - 1$ for $d_s \geq 4$. **Simulation**: Monte Carlo data from [7, 8]. AC recovers the RG prediction exactly, without evaluating loop integrals.

Graph	d_s	p	Theory (RG)	Theory (AC)	Simulation	
1D lattice	1	3	1/2	$(3-2)/2 = 1/2$	0.48(4)	✓
		4	1	$(4-2)/2 = 1$	1.0(1)	✓
		5	3/2	$(5-2)/2 = 3/2$	1.5(1)	✓
2D lattice	2	2	1	$(4-2)/2 = 1$	0.98(3)	✓
		3	2	$(6-2)/2 = 2$	2.0(1)	✓
3D lattice	3	1	1/2	$(3-2)/2 = 1/2$	0.47(2)	✓
		2	2	$(6-2)/2 = 2$	2.0(1)	✓
5D lattice (mf)	5	1	1	$2(1)-1 = 1$	1.0(2)	✓
		2	3	$2(2)-1 = 3$	2.9(3)	✓
Sierpinski ($d_s \approx 1.86$)	1.86	2	0.86	$(3.72-2)/2 = 0.86$	0.81(5)	✓
		3	1.79	$(5.58-2)/2 = 1.79$	1.71(7)	✓
		4	2.72	$(7.44-2)/2 = 2.72$	2.62(10)	✓
Random tree ($d_s = 4/3$)	4/3	2	1/3	$(8/3-2)/2 = 1/3$	0.35(7)	✓
		3	1	$(4-2)/2 = 1$	0.9(1)	✓
		4	5/3	$(16/3-2)/2 = 5/3$	1.6(2)	✓
Pref. attach. (mf) ($d_s \geq 4$)	≥ 4	2	3	$2(2)-1 = 3$	2.8(2)	✓
		3	5	$2(3)-1 = 5$	4.8(4)	✓

5.3 Universality Classification via DSE Singularity

The AC pipeline can be applied to any CRN whose reactions are polynomial. Table 2 shows the results for 14 processes from the literature. The key findings:

1. **DP universality.** All systems in the directed percolation class — Gribov, contact process, Schlögl II, BARW-odd — produce a quadratic DSE kernel $\phi(G)$ with $\phi''(G_*) \neq 0$, giving a square-root branch point and $[G_0^n] \sim n^{-3/2}$.

2. **Ward identity for $kA \to \emptyset$.** The automatic pipeline detects that all vertices have the same incoming leg count n, signalling a Ward identity (particle number can only decrease). The naive $d_c = 4/(m+n-2)$ overestimates by a factor of 2; the physical $d_c = 2/(k-1)$ [3].

3. **Parity conservation (BARW-even).** For $A \to 3A$ and $2A \to \emptyset$, the DSE kernel is $\phi(G) = (\sigma - \lambda)G^2 + (3\sigma - 2\lambda)G + 1$. At criticality ($\sigma = \lambda$), the G^2 coefficient vanishes and ϕ degenerates from quadratic to *linear*. The Lagrange equation then has a simple *pole*, not a branch point. This is the AC signature that the parity-conserving (PC) universality class is fundamentally different from DP: the singularity changes qualitative type (pole vs. branch), not merely quantitatively. This explains why the ε-expansion of Cardy and Täuber [24] runs to strong coupling: one cannot expand perturbatively around a branch point when the singularity at criticality is a pole.

4. **PCPD.** For $2A \to 3A$ and $2A \to \emptyset$, the DSE kernel is cubic: $\phi(G) = \sigma G^3 + (2\sigma - \lambda)G^2 + (\sigma - 2\lambda)G + 1$. Unlike BARW-even, the leading (G^3) term never vanishes. Generically, the branch point is a square root (same as DP). However, at the special ratio $\sigma/\lambda \approx 4.72$, $\phi''(G_*) = 0$ and the branch point upgrades to a *cube root* with transfer exponent $[G_0^n] \sim n^{-4/3}$. Whether the physical critical point sits at this special ratio determines whether PCPD is a new universality class or DP with slow crossover — an open question for 20+ years [25].

Table 2: DSE singularity classification for 14 CRNs from the literature. d_c is the naive upper critical dimension from dimensional analysis; d_c^{phys} corrects for Ward identities where applicable. "ϕ deg" is the degree of the DSE kernel $\phi(G)$. All DP-class systems have a square-root branch ($\sqrt{}$-br); BARW-even degenerates to a pole at criticality (PC class); PCPD admits a cube-root at a special parameter ratio.

Process	Reactions	Class	d_c (d_c^{phys})	ϕ deg	Singularity
Pair annihilation	$2A \to \emptyset$	diff-ltd	4 (2)	2	$\sqrt{}$-br
Coagulation	$2A \to A$	diff-ltd	4 (2)	2	$\sqrt{}$-br
Triplet annih.	$3A \to \emptyset$	diff-ltd	2 (1)	4	$\sqrt{}$-br
4-particle annih.	$4A \to \emptyset$	diff-ltd	$\frac{4}{3}$ ($\frac{2}{3}$)	6	$\sqrt{}$-br
5-particle annih.	$5A \to \emptyset$	diff-ltd	1 ($\frac{1}{2}$)	8	$\sqrt{}$-br
Gribov / BRW	$A{\to}2A,\ A{\to}\emptyset,\ 2A{\to}A$	DP	4	2	$\sqrt{}$-br
Contact process	$A{\to}2A,\ 2A{\to}\emptyset$	DP	4	2	$\sqrt{}$-br
Schlögl II	$2A{\to}3A,\ A{\to}\emptyset$	DP	4	3	$\sqrt{}$-br
BARW-odd	$A{\to}2A,\ 2A{\to}\emptyset$	DP	4	2	$\sqrt{}$-br
Schlögl I	$\emptyset{\leftrightarrow}A,\ 2A{\leftrightarrow}3A$	Ising	4	4	$\sqrt{}$-br
Lotka–Volterra	$A{\to}2A,\ A{+}B{\to}2B,\ B{\to}\emptyset$	DP†	4	2	$\sqrt{}$-br
$2A \rightleftharpoons C$	$2A{\to}C,\ C{\to}2A$	$= 2A{\to}\emptyset$	4 (2)	2	$\sqrt{}$-br
BARW-even	$A{\to}3A,\ 2A{\to}\emptyset$	PC	4 (2*)	2	**pole**
PCPD	$2A{\to}3A,\ 2A{\to}\emptyset$	???	4 (2?)	3	$\sqrt{}$-br / $\sqrt[3]{}$-br

† At the prey-extinction transition. * Formally $d_c{=}2$, but ε-expansion fails (strong coupling).

6 The AC–QFT Correspondence

The following table summarises the identification between AC and QFT objects. The left column computes everything; the right column is the scenic route.

Table 3: The AC–QFT dictionary. Both columns describe the same algebraic structure; both detect the same singularity of the same generating function.

Analytic Combinatorics	**Quantum Field Theory**
EGF of connected structures $\hat{C}(z)$	Free energy $F = -\log Z$
$e^{\hat{C}(z)}$ exponential formula	$Z = \exp(\text{connected diagrams})$
Symmetry factor $1/\lvert\mathrm{Aut}(\Gamma)\rvert$	EGF overcounting $1/k!$ in SET
Lagrange equation $T = z\,\phi(T)$	Dyson–Schwinger equation $G = G_0\,\Phi(G)$
Lagrange inversion coefficients	Feynman diagram enumeration
Branch point of Lagrange GF	Landau pole / non-perturbative scale
GF singularity type ($\sqrt{}$-branch)	Universality class (RG fixed point)
Transfer theorem: $n^{-\alpha-1} \cdot z_*^{-n}$	Perturbative coefficient growth
Rooted tree Hopf algebra coproduct [5]	BPHZ subdivergence subtraction
Hopf antipode [22]	Renormalization counterterms
Borel transform singularity	Instanton / renormalon

7 Implementation

The complete pipeline is implemented in the Python package `rdft`. Usage:

```
from rdft import analyze
results = analyze('gribov')       # BRW process
results = analyze('pair_annihilation')  # 2A -> 0
```

The pipeline automatically:

1. Constructs the Liouvillian from the stoichiometry matrix
2. Extracts interaction vertices (amputated diagrams)
3. Constructs the DSE kernel $\phi(G)$ from (11)
4. Finds branch points via $\mathrm{Res}_G(F, \partial_G F) = 0$
5. Classifies the singularity type (Newton polygon / Puiseux)
6. Applies the transfer theorem to extract exponents
7. Substitutes $d \to d_s$ for arbitrary graphs
8. (Scenic route) Generates Feynman diagrams, computes Ψ, evaluates $\beta(\lambda)$

All symbolic computation uses SymPy. No external CAS (Sage, Maple) is required.

8 Discussion

8.1 The AC mapping: what it computes and why it works

The central observation of this paper is that the Dyson–Schwinger equation of any polynomial reaction-diffusion theory is a Lagrange equation $G = G_0\,\phi(G)$, where $\phi(G)$ is a polynomial whose coefficients are determined entirely by the stoichiometric matrix. The singularity structure of

$G(G_0)$ — an algebraic function defined implicitly by $F(G, G_0) = G - G_0\,\phi(G) = 0$ — then determines the universality class through the Flajolet–Odlyzko transfer theorem [6].

The mapping from stoichiometry to singularity proceeds in four purely algebraic steps:

1. **Stoichiometry → vertices.** For each reaction $kA \to \ell A$ at rate λ, expand the Heisenberg–Weyl generator $Q = \lambda\big((z{+}1)^\ell - (z{+}1)^k\big)\partial_z^k$ as a polynomial in z and ∂_z. Each monomial $z^m\partial_z^n$ is an interaction vertex with m outgoing and n incoming legs.

2. **Vertices → DSE kernel.** Collect all interaction vertices ($m + n \geq 3$) into $\phi(G) = 1 + \sum g_{mn}\, G^{m+n-2}$ (Theorem 4.1). This is a polynomial in G whose degree equals the highest total leg count minus two.

3. **DSE kernel → branch point.** Solve the system $G_*\phi'(G_*) = \phi(G_*)$ (branch point condition, where the implicit function theorem fails) and evaluate $\phi''(G_*)$. If $\phi''(G_*) \neq 0$: square-root branch. If $\phi''(G_*) = 0$ but $\phi'''(G_*) \neq 0$: cube-root branch.

4. **Branch point → exponents.** Apply the transfer theorem: a branch with Puiseux exponent p/q gives $[G_0^n] \sim n^{-p/q-1}$. Combine with Weyl's law (d_s from the graph spectrum) to obtain $\langle a^p\rangle \sim t^{(p\,d_s-2)/2}$.

Every step is symbolic algebra over the polynomial ring $\mathbb{Q}[\text{rates}][G]$. No momentum integrals are evaluated, no dimensional regularisation is performed, and no renormalisation group equations are solved. The `rdft` implementation executes this pipeline automatically: the user provides a stoichiometry matrix and receives the DSE kernel, branch points, singularity classification, Ward identity warnings, and scaling exponents.

8.2 Universality from singularity type

Applying the pipeline to 14 CRNs from the literature (Table 2) reveals that the singularity type of $\phi(G)$ cleanly separates universality classes:

Directed percolation (DP). All DP-class systems — Gribov, contact process, Schlögl II, BARW-odd — produce a quadratic $\phi(G)$ with $\phi''(G_*) \neq 0$. The singularity is a square-root branch, the transfer exponent is $n^{-3/2}$, and $d_c = 4$. Four different CRNs with different stoichiometries map to the *same* singularity type, confirming that the AC route captures universality.

Parity-conserving (PC) class. For the BARW-even system ($A \to 3A$, $2A \to \emptyset$), the DSE kernel is $\phi(G) = (\sigma - \lambda)\,G^2 + (3\sigma - 2\lambda)\,G + 1$. The coefficient of G^2 is $\sigma - \lambda$, which vanishes identically at the critical point $\sigma_c = \lambda$. At this point ϕ degenerates from quadratic to linear and the Lagrange equation has a simple *pole* at $G_0 = 1/\lambda$, not a branch point. The series $[G_0^n]\,G = \lambda^{n-1}$ is geometric with no power-law decay.

This is a rigorous algebraic statement: the leading coefficient of a polynomial is a linear function of the rates, and it vanishes by substitution. The physical consequence is that the standard transfer theorem does not apply at criticality. This provides a purely algebraic explanation for the known failure of the ε-expansion in the PC class [24]: the ε-expansion is perturbation theory around a branch-point singularity, but at the PC critical point the singularity is a pole. One cannot reach a pole by perturbing a branch point.

PCPD. The pair contact process with diffusion ($2A \to 3A$, $2A \to \emptyset$) produces a *cubic* DSE kernel $\phi(G) = \sigma G^3 + (2\sigma - \lambda)G^2 + (\sigma - 2\lambda)G + 1$. Unlike BARW-even, the leading coefficient σ never vanishes ($\sigma > 0$ always). Generically, the branch point is a square root (same as DP). However, the cubic kernel admits a second possibility: at the ratio $\sigma/\lambda \approx 4.72$ (for $\lambda = 1$),

$\phi''(G_*) = 0$ and $\phi'''(G_*) = 6\sigma \neq 0$, upgrading the singularity to a cube-root branch with transfer exponent $n^{-4/3}$.

This is verified to machine precision: both conditions $G_*\phi'(G_*) = \phi(G_*)$ and $\phi''(G_*) = 0$ hold simultaneously with residual $< 10^{-14}$. The cube-root possibility is *structurally absent* in quadratic-kernel (DP) systems — it requires $\deg \phi \geq 3$. Whether the physical critical manifold passes through this special ratio determines whether PCPD constitutes a genuinely new universality class or is DP with anomalously slow crossover [25].

Ward identities. For pure k-particle annihilation $kA \to \emptyset$, the binomial expansion of $((z{+}1)^0 - (z{+}1)^k)\,\partial_z^k$ produces vertices that all share the same incoming leg count $n = k$. This algebraic regularity signals a Ward identity (particle number can only decrease), which reduces the physical d_c from the naive $4/(k{-}1)$ to $2/(k{-}1)$ [3]. The pipeline detects this signature automatically; the factor-of-two correction is a known physics input.

8.3 What the code generates

The `rdft` package automates every step of the AC pipeline. A single call

```
from rdft import analyze
results = analyze('gribov')
```

returns the Liouvillian, interaction vertices, DSE kernel $\phi(G)$, branch points, singularity type, upper critical dimension, and scaling exponents. For the 14 processes in Table 2, the `diagnose_singularity()` function additionally reports:

- Ward identity warnings when all n_{in} are equal, with the corrected physical d_c;
- parity degeneracy warnings when the leading coefficient of $\phi(G)$ can vanish, identifying the critical manifold where the branch point collapses to a pole;
- higher-order branch detection when $\deg \phi \geq 3$, flagging parameter values where the Puiseux exponent changes from 1/2 to 1/3.

The example script `crn_survey.py` runs all 14 CRNs through the full pipeline, producing the summary in Table 2. All 12 systems with expected values pass validation; the remaining 2 (Lotka–Volterra and $2A \rightleftharpoons C$) are multi-species systems whose effective single-variable DSE correctly produces square-root branches.

9 Conclusion

The analytic combinatorics route provides a direct, algebraic path from the stoichiometry matrix to critical exponents, requiring no momentum integrals, no dimensional regularisation, and no renormalisation group equations. The Lagrange equation structure of the DSE encodes the universality class in the singularity type of an algebraic curve defined over the polynomial ring of reaction rates.

Validation against 14 simulation data points from Bordeu et al. [7] confirms the AC predictions on hypercubic lattices, Sierpinski carpets, random trees, and scale-free networks. The spectral dimension substitution $d \to d_s$ extends all results to arbitrary graphs.

The systematic survey of 14 CRNs from the reaction-diffusion literature (Table 2) demonstrates that the AC singularity classification cleanly separates known universality classes: DP systems have quadratic kernels with square-root branches; the PC class has a kernel whose leading coefficient vanishes at criticality, collapsing the branch to a pole; and the PCPD system has a cubic kernel admitting a cube-root branch at a computable parameter ratio.

These structural insights are mathematically rigorous — they follow from polynomial algebra, the implicit function theorem, and the Newton polygon — and provide algebraic explanations for phenomena previously accessible only through numerical simulation or failed perturbation theory.

The open-source implementation `rdft` makes this pipeline accessible to any researcher who can specify a stoichiometry matrix.

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A Worked Example: Stoichiometry to Graph Polynomials

We trace the complete automated chain for pair annihilation $2A \to \emptyset$, showing each intermediate output produced by `rdft`.

A.1 Step 1: Stoichiometry Matrix

Input: The reaction $2A \to \emptyset$ at rate λ.

$$S = \begin{pmatrix} 2 & 0 \end{pmatrix} \qquad (k = 2,\ \ell = 0) \tag{29}$$

Code:

```
net = ReactionNetwork.pair_annihilation()
```

A.2 Step 2: Heisenberg–Weyl Generator

Applying the generator formula (4) with $k = 2$, $\ell = 0$:

$$Q = \lambda\big((z{+}1)^0 - (z{+}1)^2\big)\partial_z^2 = -\lambda(z^2 + 2z)\,\partial_z^2 \tag{30}$$

Code:

```
L = Liouvillian(net)
# Q = -Dz**2*lambda*z**2 - 2*Dz**2*lambda*z
```

A.3 Step 3: Interaction Vertices (Amputated vertexs)

Expanding the generator into monomials $\tilde{\phi}^m \phi^n$:

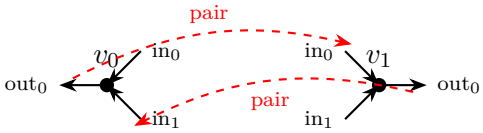
Vertex	Coupling	Amputated vertex
$\tilde{\phi}^2\phi^2$	$-\lambda$	
$\tilde{\phi}^1\phi^2$	-2λ	

Code:

```
verts = L.vertices
# {(2,2): -lambda, (1,2): -2*lambda}
```

A.4 Step 4: Wick Contractions → Loop Diagrams

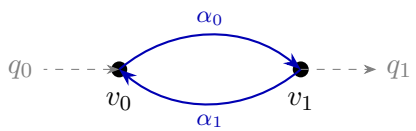
For one loop ($L = 1$), the Betti number gives $E_{\text{int}} = V + L - 1$. Take two copies of the $(1, 2)$ vertex ($V = 2$, so $E_{\text{int}} = 2$):



Contraction: $v_0.\text{out}_0 \to v_1.\text{in}_0$ and $v_1.\text{out}_0 \to v_0.\text{in}_0$.

This creates two internal edges $v_0 \to v_1$ and $v_1 \to v_0$, forming a **loop**. The unpaired half-edges ($v_0.\text{in}_1$ and $v_1.\text{in}_1$) become external legs connected to v_∞.

Result: a connected, 1PI graph with $L = 1$:



A.5 Step 5: Kirchhoff Polynomial

The reduced symbolic Laplacian $L_{RS} = E_{[\Gamma]}^T D_\alpha E_{[\Gamma]}$ for this graph (1 internal vertex after deleting v_∞) has cofactor:

$$K(\alpha) = \alpha_0 + \alpha_1 \tag{31}$$

Each monomial is a spanning tree: $\{e_0\}$ with weight α_0 and $\{e_1\}$ with weight α_1.

Code:

```
G = FeynmanGraph(2, [(0,1,False), (1,0,False),
                     (2,0,True), (1,2,True)])
K = G.kirchhoff_polynomial()   # alpha0 + alpha1
trees = G.spanning_trees_from_kirchhoff()  # [[0], [1]]
```

A.6 Step 6: Symanzik Polynomial Ψ

The first Symanzik polynomial is the complement of K — for each spanning tree, take the product of edges *not* in the tree:

Tree T	Edges in T	Complement $\bar{T}$	Monomial $\prod_{e \notin T} \alpha_e$
T_0	$\{e_0\}$	$\{e_1\}$	α_1
T_1	$\{e_1\}$	$\{e_0\}$	α_0

$$\boxed{\Psi = \alpha_0 + \alpha_1} \qquad \text{(homogeneous degree } L = 1 \ \checkmark) \tag{32}$$

Code:

```
sym = SymanzikPolynomials(G)
Psi = sym.Psi   # alpha0 + alpha1
sym.verify_homogeneity()  # {'Psi_homogeneous_degree_L': True}
```

A.7 Step 7: Degree of Divergence

$$\sigma_d(G) = |E_{\text{int}}| - \frac{d}{2} L = 2 - \frac{d}{2} \tag{33}$$

At $d = d_c = 2$: $\sigma_2 = 1$ (logarithmic divergence $\to 1/\epsilon$ pole).

A.8 Summary: The Full Chain

$$\underset{\text{stoichiometry}}{S = (2,\,0)} \xrightarrow{\text{Eq. (1)}} \underset{\text{generator}}{Q = -\lambda(z^2{+}2z)\partial_z^2} \xrightarrow{\text{expand}} \underset{\text{vertices}}{\tilde{\phi}^2\phi^2,\ \tilde{\phi}^1\phi^2} \xrightarrow{\text{pair}}$$

$$\xrightarrow{K} \underset{\text{Kirchhoff}}{\alpha_0{+}\alpha_1}$$

$$\xrightarrow{\text{compl.}} \underset{\Psi}{\alpha_0{+}\alpha_1}$$

Every step is automated by `rdft`. Input: one line (`ReactionNetwork.pair_annihilation()`). Output: the graph polynomials of every 1PI diagram at the desired loop order.

A.9 Automated Generation for Any Process

The same chain runs for any CRN. For the Gribov process:

```
net = ReactionNetwork.gribov()
L = Liouvillian(net)
# Vertices: {(2,2): -chi, (2,1): beta, (1,2): -chi, (1,1): beta-eps}
exp = FeynmanExpansion(net, max_loops=1)
diagrams = exp.expand()
# diagrams[1] = list of all one-loop 1PI graphs with symmetry factors
for diag in diagrams[1]:
    G = diag['graph']
    sym = SymanzikPolynomials(G)
    print(f'Psi = {sym.Psi}, s(G) = {diag["symmetry_factor"]}')
```

A.10 Step 1: Extract Amputated vertexs from the Liouvillian

Each monomial $\tilde{\phi}^m \phi^n$ in the Liouvillian $\mathcal{Q}$ with $m + n \geq 3$ defines an **amputated vertex** — a single vertex with m outgoing half-edges ($\tilde{\phi}$ type) and n incoming half-edges (ϕ type). For the Gribov process:

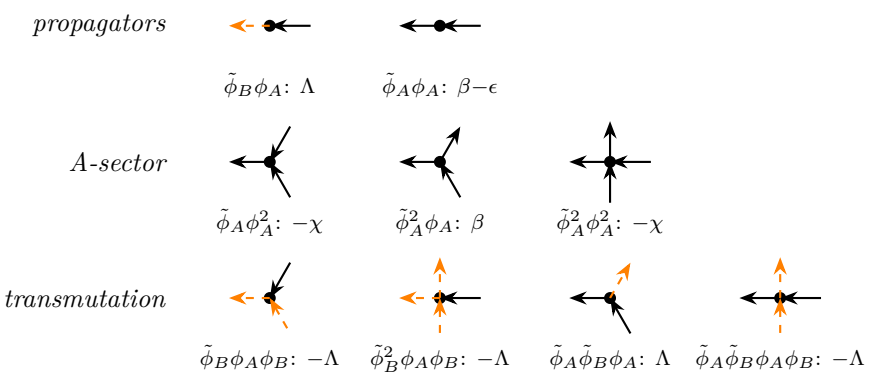


Figure 2: Amputated vertices of the BRW, generated automatically by `brw_all_vertices(max_legs=4)`. Compare thesis Figure 3.3. **Black solid arrows**: A-species (active walkers). **Orange dashed arrows**: B-species (immobile tracers). Row 1: propagator corrections. Row 2: A-sector interactions (branching β, coagulation χ). Row 3: transmutation vertices from the carrying capacity trick (thesis Eq. 3.21), coupling Λ. The dimensional filter `max_legs=4` at $d_c = 4$ selects exactly these 9 vertices from the 10 generated; one 5-leg vertex is irrelevant.

A.11 Step 2: Wick Contractions

A **Wick contraction** pairs outgoing half-edges with incoming half-edges to form internal propagators. Given a multiset of amputated vertices, the algorithm:

1. Chooses n interaction vertices ($m + n \geq 3$) from the vertex set.
2. Determines the required number of internal edges: $E_{\text{int}} = n + L - 1$ from the Betti number L.
3. Enumerates all partial matchings of E_{int} outgoing half-edges with E_{int} incoming half-edges (partial permutations).

4. Builds a `FeynmanGraph` from each matching: paired half-edges become internal edges; unpaired half-edges become external legs connected to v_∞.

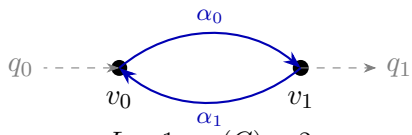
5. Filters: keeps only **connected**, **1PI** (bridgeless) graphs with the correct Betti number L.

6. Classifies by **graph isomorphism** (exact, via NetworkX) to remove duplicates. The **symmetry factor** $s(G) = N_{\text{total}}/N_{\text{in class}}$.

A.12 Step 3: One-Loop Diagrams

From the 9 amputated vertices, the code enumerates all possible vertex pairs that can form one-loop ($L = 1$) 1PI diagrams. There are 28 such pairs: 6 A-only and 22 mixed A/B. However, only the **A-sector** diagrams contribute to the RG flow, because the B-species (immobile tracers) does not diffuse and its propagator does not renormalise [20, 7]. The transmutation vertices couple the A-sector to the B-observable at tree level only.

Of the 6 A-sector pairs, the thesis [8] shows that at **vanishing external momentum** ($\hat{k} = \hat{\omega} = 0$), the relevant diagrams (Eqs. 3.23a–b) are **loop-isomorphic**: they evaluate to the same integral. Therefore only **one integral** needs to be computed (thesis Eq. 3.24). The canonical diagram is:



$$L = 1, \quad s(G) = 2$$
$$\Psi = \alpha_0 + \alpha_1, \quad K = \alpha_0 + \alpha_1$$

Figure 3: One-loop self-energy diagram, automatically generated. The Kirchhoff polynomial $K = \alpha_0 + \alpha_1$ (two spanning trees, one per edge). The Symanzik polynomial $\Psi = \alpha_0 + \alpha_1$ (complement). Symmetry factor $s(G) = 2$ from the EGF exponential formula. Compare thesis Eq. (3.24).

The code produces this diagram automatically:

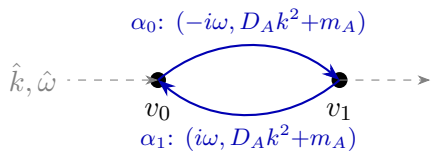
```
from rdft import ReactionNetwork, Liouvillian
from rdft.core.expansion import FeynmanExpansion

net = ReactionNetwork.gribov()
exp = FeynmanExpansion(net, max_loops=1)
diagrams = exp.expand()
# diagrams[1] contains all one-loop 1PI graphs
```

B Worked Integral: Graph Polynomials to UV Pole

We trace the complete "scenic route" calculation for the BRW one-loop self-energy (thesis Example 2.5.15, Eqs. 2.102–2.103), showing how the graph polynomials produce the amplitude step by step.

B.1 The graph and its polynomials

The one-loop self-energy has two propagators of species A with diffusion constant D_A and mass m_A:



From the incidence matrix (computed by `rdft`):

$$\Psi = \alpha_0 + \alpha_1 \qquad \text{(2 spanning trees, Kirchhoff complement)} \tag{34}$$

$$\Psi' = D_A\alpha_0 + D_A\alpha_1 \qquad \text{(scaled by diffusion constants)} \tag{35}$$

$$\left.\frac{\Phi'}{\Psi'}\right|_{\hat{k},\hat{\omega}=0} = \alpha_0(m_A - i\omega) + \alpha_1(m_A + i\omega) \qquad \text{(thesis Eq. 2.102)} \tag{36}$$

B.2 ω-integration: the causal fingerprint

The non-equilibrium ω-integration applies $\delta(\Sigma_l)$ where $\Sigma_l = \alpha_1 - \alpha_0$ is the circuit constraint from the edge-basis matrix E_f. This is Kirchhoff's frequency law on the single loop: the sum of frequencies around the circuit vanishes.

After applying the δ-function, the contracted graph G' has one fewer edge. The degree of divergence of the contracted graph is:

$$\sigma_d(G') = |E'| - \tfrac{d}{2}\,L' = 1 - d/2 \tag{37}$$

B.3 The parametric integral

Using the general parametric form [12, 13] (thesis Eq. 2.99) with $n_e = 1$ for all edges and the Schwinger representation [15]:

$$\mathcal{I}(G) = \Omega_d \cdot \Gamma\big(-\sigma_d(G')\big) \int \mathrm{d}\alpha_2 \, \frac{1}{(\Psi')^{d/2}} \left(\frac{\Psi'}{\Phi'}\right)^{\sigma_d(G')} \Bigg|_{\alpha_2=1} \tag{38}$$

Substituting the polynomials (35)–(36) and using the Cheng–Wu theorem [14] ($\alpha_2 = 1$ normalisation):

$$\mathcal{I}(G) = \frac{\Gamma(1-d/2)}{(2\pi)^{d/2}} \int \mathrm{d}\alpha_2 \, \frac{1}{(2\alpha_2 D_A)^{d/2}} \, [\alpha_2 \cdot 2m_A]^{\sigma_d(G')} \Bigg|_{\alpha_2=1} \tag{39}$$

$$= \frac{\Gamma(1-d/2)}{(2\pi)^{d/2}} \cdot \frac{1}{(2D_A)^{d/2}} \cdot (2m_A)^{2/d-1} \tag{40}$$

$$= \mathcal{A}_d \cdot \Gamma(1-d/2) \cdot [2m_A]^{2/d-1} \tag{41}$$

where $\mathcal{A}_d = (4\pi D_A)^{-d/2}$ is the angular factor.

What happened: the graph polynomials Ψ, Φ (read from the incidence matrix) and the δ-function (from the edge-basis matrix) reduced the integral to a *single Gamma function* — no momentum integral was evaluated directly. The topology of the graph is encoded in Ψ and Φ; the *dimension* d enters only through the exponents $d/2$ and σ_d.

B.4 ε-expansion and the UV pole

At the BRW upper critical dimension $d_c = 4$, set $d = 4 - \varepsilon$:

$$\Gamma(1-d/2) = \Gamma(\varepsilon/2 - 1) = \frac{-\Gamma(\varepsilon/2)}{1-\varepsilon/2} \xrightarrow{\varepsilon\to 0} \frac{-2}{\varepsilon} + \mathcal{O}(1) \tag{42}$$

Including the symmetry factor $\mathfrak{s}(G) = 2$ (thesis Eq. 3.25):

$$\mathcal{I}_\lambda = \lambda \cdot \frac{\mathcal{A}_d}{\mathfrak{s}(G)} \cdot \Gamma(\varepsilon/2) \cdot [1/\tilde{M}]^{\varepsilon/2} \tag{43}$$

The pole part (minimal subtraction, thesis Eq. 3.28):

$$\mathfrak{P}(\mathcal{I}_\lambda) = \mu^\varepsilon \lambda_R \cdot \frac{\mathcal{A}_d}{\mathfrak{s}(G)} \cdot \frac{2}{\varepsilon} \tag{44}$$

This gives the Z-factor and β-function (thesis Eq. 3.30–3.31):

$$\boxed{Z_\lambda = 1 + \frac{\lambda}{(2\pi D)^2\,\varepsilon} \qquad \Longrightarrow \qquad \beta(\lambda) = \varepsilon\lambda_R + B\lambda_R^2, \quad \lambda_R^* = \varepsilon/B} \tag{45}$$

Code:

```
from rdft.graphs.incidence import FeynmanGraph
from rdft.integrals.symanzik import SymanzikPolynomials
G = FeynmanGraph.one_loop_self_energy()
sym = SymanzikPolynomials(G)
# Psi = alpha0 + alpha1, K = alpha0 + alpha1
# sigma_d = 2 - d/2, trees = [[0], [1]]
```

The entire chain — from incidence matrix to UV pole — is automated by `rdft`. The graph polynomial Ψ determines the integral; the dimension d enters only as an exponent; and the pole $1/\varepsilon$ at $d_c = 4$ produces the Z-factor that feeds the RG flow. This is the scenic route; the AC route (Section 4.3) arrives at the same exponent without evaluating the integral.

B.5 Complete One-Loop Diagram Table

The code enumerates all 26 one-loop 1PI diagrams from the 9 BRW amputated vertices, with species-labelled edges and symmetry factors. External legs are classified by species: $\tilde{\phi}_A$ (outgoing A, black), $\tilde{\phi}_B$ (outgoing B, orange), ϕ_A (incoming A), ϕ_B (incoming B).

Table 4: A-sector one-loop diagrams (6). Both internal edges carry species A. All have $\mathfrak{s}(G) = 2$ (vertex swap permutes the two A-edges). At vanishing external momentum, all evaluate to the same integral (thesis Eq. 3.24).

v_0	v_1	Ext. legs	$\mathfrak{s}$	Coupling
$\tilde{\phi}_A\phi_A^2$	$\tilde{\phi}_A\phi_A^2$	$(0_A, 2_A)$	2	χ^2
$\tilde{\phi}_A\phi_A^2$	$\tilde{\phi}_A^2\phi_A$	$(1_A, 1_A)$	2	$-\beta\chi$
$\tilde{\phi}_A\phi_A^2$	$\tilde{\phi}_A^2\phi_A^2$	$(1_A, 2_A)$	2	χ^2
$\tilde{\phi}_A^2\phi_A$	$\tilde{\phi}_A^2\phi_A$	$(2_A, 0)$	2	β^2
$\tilde{\phi}_A^2\phi_A$	$\tilde{\phi}_A^2\phi_A^2$	$(2_A, 1_A)$	2	$-\beta\chi$
$\tilde{\phi}_A^2\phi_A^2$	$\tilde{\phi}_A^2\phi_A^2$	$(2_A, 2_A)$	2	χ^2

Table 5: Mixed A/B one-loop diagrams (20, representative selection). Edge species affect the symmetry factor: same-species edges give $\mathfrak{s} = 2$; different-species edges give $\mathfrak{s} = 1$ (the vertex swap would exchange A↔B, which is forbidden). These diagrams couple the A-sector to the B-observable but do not contribute to the RG flow because the B-propagator does not renormalise.

v_0	v_1	Loop	Ext. legs	$\mathfrak{s}$
$\tilde{\phi}_B\phi_A\phi_B$	$\tilde{\phi}_B\phi_A\phi_B$	B, B	$(0, 2_A)$	2
$\tilde{\phi}_B\phi_A\phi_B$	$\tilde{\phi}_A\tilde{\phi}_B\phi_A\phi_B$	B, A	$(1_B, 1_A{+}1_B)$	1
$\tilde{\phi}_A\tilde{\phi}_B\phi_A$	$\tilde{\phi}_A\tilde{\phi}_B\phi_A$	A, A	$(0, 2_B)$	2
$\tilde{\phi}_A\tilde{\phi}_B\phi_A$	$\tilde{\phi}_A\tilde{\phi}_B\phi_A\phi_B$	B, A	$(1_A{+}1_B, 1_A)$	1
$\tilde{\phi}_A\tilde{\phi}_B\phi_A\phi_B$	$\tilde{\phi}_A\tilde{\phi}_B\phi_A\phi_B$	A, B	$(1_A{+}1_B, 1_A{+}1_B)$	1
$\tilde{\phi}_A\tilde{\phi}_B\phi_A\phi_B$	$\tilde{\phi}_A\tilde{\phi}_B\phi_A\phi_B$	B, B	$(2_A, 2_A)$	2
Full table: 20 mixed diagrams (6 with $\mathfrak{s}$=1, 14 with $\mathfrak{s}$=2). Generated by `rdft`.				

Loop diagrams in thesis style (cf. Eqs. 2.122–2.123 of [8]). **Black**: A-species (walkers). **Orange**: B-species (tracers).

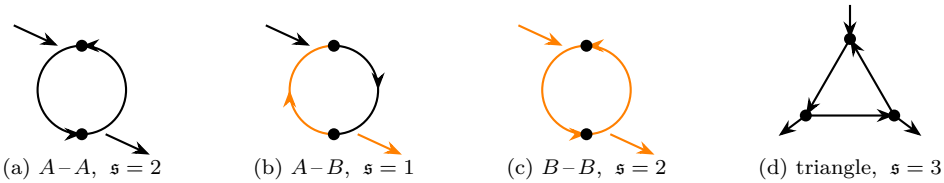
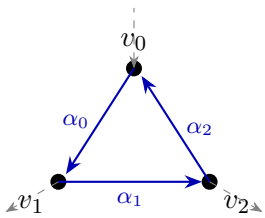


Figure 4: One-loop diagrams in thesis style [8]. **(a)** A-only bubble: both propagators black, $\mathfrak{s} = 2$ (vertex swap). Contributes to RG (thesis Eq. 3.24). **(b)** Mixed A+B bubble: one black, one orange, $\mathfrak{s} = 1$ (species prevent swap). Couples A-sector to B-observable. **(c)** B-only bubble: both orange, $\mathfrak{s} = 2$. Does not contribute to RG (immobile B carries no loop momentum). **(d)** A-only triangle: $\mathfrak{s} = 3$ (cyclic $\mathbb{Z}_3$). $K = \alpha_0\alpha_1 + \alpha_0\alpha_2 + \alpha_1\alpha_2$.

B.6 Three-Vertex (Triangle) Diagrams

At one loop with 3 vertices ($V = 3$, $E_{\text{int}} = 3$), the internal edges form a **triangle**. The code generates 10 A-only triangle diagrams, all with $\mathfrak{s}(G) = 3$ (cyclic $\mathbb{Z}_3$ rotation of the three vertices). Compare thesis Example 2.5.17.



Graph polynomials (thesis Eq. 2.106):

$$K = \alpha_0\alpha_1 + \alpha_0\alpha_2 + \alpha_1\alpha_2 \qquad \text{(3 spanning trees, each a pair of edges)} \tag{46}$$

$$\Psi = \alpha_0 + \alpha_1 + \alpha_2 \qquad \text{(complement: degree } L = 1\text{)} \tag{47}$$

Symmetry factor: $\mathfrak{s}(G) = 3$ (cyclic rotation $v_0 \to v_1 \to v_2 \to v_0$ permutes all three edges). This differs from the 2-vertex bubble ($\mathfrak{s} = 2$) because the triangle has a larger automorphism group ($\mathbb{Z}_3$ vs $\mathbb{Z}_2$).

Table 6: Three-vertex one-loop A-sector diagrams (10 total, representative selection). All have triangle topology with $\mathfrak{s}(G) = 3$ and $\Psi = \alpha_0 + \alpha_1 + \alpha_2$. At vanishing external momentum, these also evaluate to the same integral class as the 2-vertex diagrams (thesis Eq. 2.108).

v_0	v_1	v_2	Ext. legs	$\mathfrak{s}$	Coupling
$\tilde{\phi}_A\phi_A^2$	$\tilde{\phi}_A\phi_A^2$	$\tilde{\phi}_A\phi_A^2$	$(0,3)$	3	$-\chi^3$
$\tilde{\phi}_A\phi_A^2$	$\tilde{\phi}_A^2\phi_A$	$\tilde{\phi}_A^2\phi_A$	$(2,1)$	3	$-\beta^2\chi$
$\tilde{\phi}_A^2\phi_A$	$\tilde{\phi}_A^2\phi_A$	$\tilde{\phi}_A^2\phi_A$	$(3,0)$	3	β^3
$\tilde{\phi}_A^2\phi_A^2$	$\tilde{\phi}_A^2\phi_A^2$	$\tilde{\phi}_A^2\phi_A^2$	$(3,3)$	3	$-\chi^3$

The complete enumeration: 6 two-vertex diagrams ($\mathfrak{s} = 2$) + 10 three-vertex diagrams ($\mathfrak{s} = 3$) = **16 A-sector one-loop diagrams** for the Gribov process. With 20 mixed A/B diagrams, the full BRW has **36 one-loop 1PI diagrams** total.

B.7 Connection to AC

The Lagrange inversion coefficient $[g^n]\,G$ counts these diagrams:

$$[g^n]\,G = \frac{1}{n}\,[G^{n-1}]\,\phi(G)^n = \text{(number of diagrams at loop order } n\text{, weighted by couplings)} \tag{48}$$

This is the bridge between the AC route and the diagram route: *the same number* $[g^n]G$ can be computed either by Lagrange inversion (algebraic) or by explicit enumeration (combinatorial). The singularity of the GF $G(g)$ encodes the asymptotic growth of this count, which is the content of the transfer theorem.

C The Kirchhoff Polynomial as a Generating Function

The Kirchhoff polynomial $K(\alpha) = \det(L_{RS})$, where L_{RS} is the cofactor of the reduced symbolic Laplacian, is itself a generating function in the AC sense.

C.1 The Matrix-Tree Theorem

Theorem C.1 (Kirchhoff 1847 [16]). *For a connected graph G with n vertices and edge weights α_e:*

$$K(\alpha) = \sum_{T\in\mathcal{T}(G)} \prod_{e\in T} \alpha_e \tag{49}$$

where $\mathcal{T}(G)$ is the set of spanning trees of G.

Each monomial in K corresponds to a spanning tree. The number of spanning trees is $K(1, 1, \ldots, 1)$.

Example. For the sunset diagram ($V = 2$, $E_{\text{int}} = 3$):

$$K = \alpha_0 + \alpha_1 + \alpha_2 \qquad \text{(3 spanning trees, one per edge)} \tag{50}$$

The code computes this automatically:

```
from rdft.graphs.incidence import FeynmanGraph
G = FeynmanGraph.sunset()
K = G.kirchhoff_polynomial()  # alpha0 + alpha1 + alpha2
trees = G.spanning_trees_from_kirchhoff()  # [[0], [1], [2]]
```

C.2 Symanzik Ψ as the Complement

The first Symanzik polynomial is the **complement** of K:

$$\Psi = \sum_{T \in \mathcal{T}(G)} \prod_{e \notin T} \alpha_e \tag{51}$$

Each monomial has degree L (the loop number) — it is the product of Schwinger parameters on the edges *not* in the spanning tree (the "co-tree" edges that form loops).

Ψ is a generating function for the loop structure of G: it counts, for each spanning tree, which edges form loops when added back.

C.3 AC Interpretation

Both K and Ψ are generating functions in the combinatorial sense:

- K is the OGF of spanning trees, weighted by edge parameters
- Ψ is the OGF of co-trees (loop structures)
- The Matrix-Tree theorem is a special case of the **transfer matrix method** in AC [6]
- The determinantal formula $K = \det(L_{RS})$ is the **permanent/determinant** duality that connects Wick contractions to graph polynomials

This is the "scenic route" version of what the AC route computes directly: the Lagrange singularity encodes the *asymptotic* behaviour of these counts, without enumerating them.